

# Financial Mathematics

## (MATH35400/MATHM5400)

Problem Class 7 – 28 February 2026 <sup>1</sup>

### Problems:

1. (2020 exam, Q3 (b)) A *down-and-out* call with knockout price  $b$  and rebate  $R$  is an ordinary American call with strike price  $K$  as long as the stock price does not fall below the knockout price. If the stock price falls below the knockout price at a random time point  $\tau < T$ , then the option expires and the owner of the option receives at time point  $\tau$  the rebate  $R$ .

Consider now a Cox-Ross-Rubinstein model with  $T = 3$  periods and stock price process  $(S_t)$  where  $S_0 = 80$ ,  $u = 3/2$ ,  $d = 1/2$  and interest rate  $r = 0.1$ .

- (a) Calculate the risk-neutral probability and the stock prices at each node in the tree.
- (b) Calculate the value of the down-and-out call with knockout price  $b = 70$ , rebate  $R = 2$  and exercise price  $K = 120$  at every time point  $t$  and for every state  $\omega$ . Then calculate a replicating strategy.
- (c) Consider a usual European call option maturing at time 3 with exercise price  $K = 120$ . Calculate the time 0 price and compare with the price of the down-and-out call.

---

<sup>1</sup>©University of Bristol 2026. This material is copyright of the University of Bristol unless explicitly stated otherwise. It is provided exclusively for educational purposes at the University and is to be downloaded or copied for your private study only.